

## Research paper

## A quantitative assessment of EOG eye tracking during free viewing in sighted and in congenitally blind

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## ABSTRACT

Electrooculography (EOG) has long been a reliable tool for detecting blinks and eye movements. However, its application in naturalistic and unrestricted settings has been little explored. We compared the accuracy of EOG-based eye tracking to a video-based tracking system during free viewing with unrestricted head and eye movements, in both normal lighting and complete darkness. Our results show that, with high signal-to-noise data, EOG reliably detects blinks and saccades, comparable to video-based tracking in light and in darkness. Applying our algorithms to EOG data recorded from blind participants or in sighted during eyes closed, eye movements are still effectively identified. In both participant groups, detected blinks and saccades retain characteristic movement profiles such as blink rate and saccade main sequence, though other features like blink length, preferred saccade direction, saccade rate, and saccade amplitude differ. Interestingly, these differences closely align between sighted individuals with eyes closed and blind participants, while remaining distinguishable from surrogate data, highlighting their genuine physiological basis. These findings affirm EOG's robustness as a tool for eye movement and blink analysis in both controlled and naturalistic environments. Our work further emphasizes the presence of blinks and saccadic muscular movements both in sighted individuals with eyes closed and in the congenitally blind.

## 1. Introduction

For decades, electrooculography (EOG) has been a valuable tool in detecting blinks and eye movements in lab-based experiments. This is due to its high reliability, high sampling rate, low cost, and easy application, making it well-suited for capturing ocular activity (Jacob & Karn, 2003; Young & Sheena, 1975). The accuracy has been found to be comparable to many video-based eye tracking devices (Jia & Tyler, 2019). Recently, there has been increased interest in gaze patterns and blink behavior as potential markers for neurodegenerative and developmental disorders such as Parkinson's disease, autism, mild cognitive impairment, and Alzheimer's disease, where blink rate and gaze irregularities are often observed (Bologna, Fasano, Modugno, Fabbri, & Berardelli, 2012; Minissi, Chicchi Giglioli, Mantovani, & Alcaniz Raya,

2022; Opwonya et al., 2022; Seligman & Giovannetti, 2015). Additionally, there is a growing body of research on the importance of naturalistic behavior in recording behavioral and neural markers, as experimental settings demanding strict movement control may not fully capture the ecological validity of certain behaviors (Cao, Chen, & Haendel, 2020; Cao & Händel, 2019; Chen, Cao, & Haendel, 2022; Gramann, Ferris, Gwin, & Makeig, 2014; Ladouce, Donaldson, Dudchenko, & Ietswaart, 2019). Accordingly, traditional EOG methods should be assessed within this new context, testing its reliability and accuracy in settings where natural movement is allowed.

We set out to assess the reliability of saccade and eyeblink detection using EOG during free viewing and without any restriction on the state of the head or the eye. To this end, we compared in a free moving setup the saccade and blink detection of EOG-based eye tracking with a video-

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based system, which is independent of electrophysiological movement artifacts in normal lighting conditions and in complete darkness. Additionally, we assessed saccade- and blink-related EOG activity in a free moving setup in congenitally blind and in sighted individuals with eyes closed. We compared those EOG measures to measures in sighted subjects with eyes open.

We tested the following hypothesis:

- EOG is functional without fixation constraints, and therefore usable under free viewing conditions and outside the lab.
- Blinks and saccadic eye movements can be detected with EOG during eyes closed states, thus making them accessible during sleep, coma, and comparable conditions.
- Using EOG, eye movements can be assessed in blind individuals for whom video-based tracking methods fail due to pupil abnormalities or partial control over the eyelids.

Given sufficiently high data quality, we found that blink rate, saccade rate, saccade amplitude and main sequences are comparable between EOG and video-based eye tracking during light and darkness in sighted individuals. When assessing eye movements during eyes closed states and in blindness, we found that blinks and saccades can still be detected and that the detected events share movement-specific features with those eye movements found in sighted with eyes open, such as the typical main sequence and velocity profile. Other characteristics, such as preferred saccade direction, rate and amplitude, as well as blink rate and length, are different between sighted with eyes-open and sighted with eyes closed or blind individuals. Interestingly, these characteristics are largely comparable between sighted individuals with eyes closed and blind individuals.

## 2. Methods

We analyzed 3 different experiments that were conducted at different sites and with different participants. Experiment 1 was conducted in Würzburg, Germany, while experiment 2 and 3 was conducted in Jerusalem, Israel. The equipment used is described in detail for each experiment separately.

Experiment 1: sighted in light and dark, eyes open.

### 2.1. Equipment

**Video-based eye tracker:** Binocular eye movements were recorded with 120 Hz sampling rate using the SMI eye-tracking glasses (ETG 2 W Analysis Pro- 120 Hz). This is an eye tracker using the principle of dark pupil tracking and pupil/ corneal reflex analysis. Near-infrared light is shone onto the eyes creating a strong contrast between the dark pupil and the surrounding lighter iris. The infrared light further reflects off the cornea creating a corneal reflection (or first Purkinje reflection). The center of the dark pupil is used to approximate the gaze direction, while the corneal reflection is a stable reference with which a correction for movements of the head with respect to the camera is possible. Subjects were naïve to the exact purpose of the eye-tracking but were explicitly told that they could move their eyes freely.

**EOG:** EOG data were collected using a Smarting mobile EEG system (mBrainTrain LLC, Serbia), with 8 recording channels and a sampling rate of 500 Hz. Three electrodes were placed per eye: below and above the eye, and to the outer canthus. A reference electrode was placed on each earlobe for possible re-referencing.

### 2.2. Participants and procedure

We measured 19 participants (13 females; mean age: 28.5 years, SD = 8.6). All had normal vision, received payments or study credits for their participation and gave their written informed consent according to the Declaration of Helsinki and the local data protection law. The study

was approved by the local ethics committee (Department of Psychology, University Würzburg). Four participants had to be excluded from further analysis due to corrupted file structure, which made it impossible to load the data.

Individuals sat comfortably in a dimly lit room without any task. The light was then turned off for 190 s and back on for 190 s. This process was repeated consecutively 5 times, resulting in 5 iterations per lighting condition. The order was randomized between participants. At the end of each testing block, the experimenter entered the testing room to turn the light on or off, which led to a short break in the range of 10 to 20 s in between blocks. For analysis, the 190-second recordings were used.

Experiment 2: sighted with eyes closed.

### 2.3. Equipment

EOG data were recorded using a g.HIamp amplifier (g.tec medical engineering, Austria) with a sampling rate of 512 Hz. We recorded from four passive electrodes (g.tec medical engineering, Austria): two electrodes were placed at the outer canthi, and two additional electrodes were placed above and below the left eye. Experimental triggers were delivered via Psychtoolbox in Matlab and were time-aligned with the EOG signals and stored for offline-analysis using a Simulink model (MathWorks, MA).

### 2.4. Participants and procedure

We measured 38 participants (28 females, mean age = 22.9, SD = 2.6). All had normal or corrected to normal vision, received payment or study credits for their participation and gave their written informed consent. All procedures were approved by the institutional review board of ethical conduct (The Hebrew University of Jerusalem). The data of four participants had to be excluded from further analysis due to excessive noise in the recording.

Participants sat comfortably in a dimly lit room. They were instructed to close their eyes and relax for two minutes; accordingly, recordings were 2 min long. The first and last 100 samples of each recording (0.195 s) were discarded from further analysis.

Experiment 3: congenitally blind.

### 2.5. Equipment

EOG data were recorded using a 64-channel whole-head BioSemi EEG system with a sampling rate of 256 Hz.

### 2.6. Participants and procedure

We measured 8 congenitally blind participants (6 females, mean age = 34.5, SD = 9.27). Participants were recruited based on their self-reported clinical history. All participants lacked residual light perception but one who reported faint light perception. All participants had pre- or perinatal anomalies in the eyes, resulting from retinopathy of prematurity (N = 4), microphthalmia (N = 1), microphthalmia and retinal detachment (N = 1), Leber congenital amaurosis (N = 1) or congenital rubella (N = 1). All procedures were approved by the institutional review board of ethical conduct (The Hebrew University of Jerusalem).

Participants sat comfortably in a dimly lit room. There were 4 EEG blocks. In each block, participants listened to soundscapes (30 total, 1.5 sec long each), each repeated 8 times, leading to a total of 240 soundscapes. The soundscapes were chosen from the “The vOICe” (Meijer, 1992), a visual-to-auditory sensory substitution device that converts visual images to sound representations using a pre-defined algorithm. The soundscapes used consisted of letters, faces, and scrambled faces on which participants had previously been trained. Participants were instructed to identify the presented soundscape (i.e., in their head), but no actual response was required. For the purpose of this analysis, the

EOG data was analyzed independently of the soundscapes. The recording lasted around 15 min. The first 500 samples (1.953 s) and the last 200 samples (0.781 s) of each recording were discarded from further analysis.

Table 1 provides an overview of the 3 experimental setups. Analysis.

## 2.7. EOG

### 2.7.1. Noise and artifact rejection

For artifact and noise detection, we created a vertical EOG signal (vEOG), by subtracting the data recorded by the electrode above the eye from the data recorded by the electrode below the eye (Schlögl et al., 2007). Dependent on the setup, this resulted in one or two vEOG data streams. The horizontal EOG (hEOG) was created by subtracting the data recorded by the electrode that was placed on the outer canthi of the left eye from the data recorded by the electrode that was placed on the outer canthi of the right eye. This data was subject to visual inspection to detect jumps and artifacts due to strong external noise and gross movement artifacts. Detected artifacts were removed by replacing data with NaN's (not a number). Following this, an automatic artifact rejection and noise reduction routine was applied to each electrode's raw data: 1) If 3 consecutive values were identical, the values were set to NaN  $\pm$  50 samples around the event. 2) Data was linearly interpolated in order to apply a filter; however, all NaNs were again included after filtering for further analysis. 3) The data was filtered with a 7 Hz high pass filter (Butterworth, zero-phase, filter order = 2) and a 48 – 52 Hz notch filter (Butterworth, zero-phase, filter order = 6) to reduce line noise. 4) Values above 20 STD ( $\pm$  5 samples) and solitary values (single values surrounded by NaNs) were replaced by NaNs.

### 2.7.2. Blink detection using EOG

After artifact rejection and filter application, we calculated the vEOG. For Experiment 1, this was done for the two eyes separately. For blink detection, we calculated the z-value of each vEOG sample by subtracting the mean and dividing by the standard deviation of the entire vEOG signal. For experiment 1 this was done per block. A blink was defined by z-values below a threshold of 0 (z-values below 0 were combined if closer than 10 ms) with an additional z-value below  $-2$  in this time window. A blink was only assumed if the time period was longer than 75 ms with a maximum length of 500 ms. Additionally, if a z-value in the time between 100 ms before or after the blink onset or offset was greater than 75 % of the absolute minimum during the blink, the event was not considered a blink as this shape is rather found for large saccades. Blink start was defined by the last value above threshold, blink end was defined by the last value still below the threshold. Blink rate was calculated as blinks per minute. Only artifact-free vEOG was included in the blink rate calculation. For experiment 1, a binocular blink was registered if a left eye blink and right eye blink overlapped in

time. Blink timing was defined by the overlap.

### 2.7.3. Saccade detection using EOG

Saccadic events were detected based on eye-movement velocity, a method developed by Engbert and Kliegl (2003). After artifact rejection and filter application, an elliptic threshold with a radius around the root mean square of the velocity values was applied. The threshold was set to 2 times the root mean square of the velocity values. The saccade velocity for saccade detection was calculated following the approach by (Engbert & Kliegl, 2003) using 5 data points. This approach is more robust and less prone to measurement noise. Please note that for the EOG data used for saccade detection, an additional 25 Hz low pass filter was applied (Butterworth, zero-phase, filter order = 4). If the squared sum of the vEOG and the hEOG velocity value exceeded the threshold in two consecutive samples, the event was marked as a possible saccade.

The following criteria determined if the event was indeed considered a saccade, giving special consideration to the temporal succession of events: 1) If two events were detected within a 50 ms window yet occurred towards opposite directions (e.g. an event towards the left and an event towards the right), they were discarded unless the velocity value of one of the events was more than double in size compared to the other event. 2) The remaining events that were less than 15 ms apart were combined to a single event. 3) Only events lasting more than 16 ms and shorter than 200 ms were considered as saccades.

Saccade start was defined by the first value of an event above half the threshold, saccade end was defined by the last value still above half the threshold. Saccade rate was calculated as saccades per second. The saccade amplitude was based on the Euclidian distance of the vertical and horizontal amplitude. Specifically, we calculated the amplitude of a given saccade based on the maximum in the vertical and horizontal direction during the saccade. The velocity calculation was based on the first derivative of a 5-sample time window, i.e. the mean of the first 2 samples minus the mean of the last 2 samples within this time window (Engbert & Kliegl, 2003).

A binocular saccade was assumed if a left eye saccade and a right eye saccade coincided (only relevant for experiment 1). Saccade time was defined by the time of overlap. Horizontal and vertical saccade amplitude was calculated as the mean of the maximum amplitude over both eyes within the binocular time window.

### 2.7.4. Velocity signal for visualization

The velocity signal for visualization was calculated as the first derivation of the Euclidian distance, i.e. the root of the sum of the square of the horizontal position and the square of the vertical position (averaged over the left and right eye for experiment 1). The velocity was then low pass filtered at 25 Hz (Butterworth, zero-phase, filter order = 4). Note that the velocity signal used for saccade detection was calculated in a more robust way (see Methods: Saccade detection using EOG).

**Table 1**

The experimental setups of the three studies are summarized.

|                               | Experiment 1                                                         | Experiment 2                                       | Experiment 3                                                     |
|-------------------------------|----------------------------------------------------------------------|----------------------------------------------------|------------------------------------------------------------------|
| <b>Location</b>               | Würzburg, Germany                                                    | Jerusalem, Israel                                  | Jerusalem, Israel                                                |
| <b>Participant Group</b>      | Sighted (eyes open, light/dark conditions)                           | Sighted (eyes closed)                              | Congenitally blind                                               |
| <b>Number of Participants</b> | 19 (16 analyzed)                                                     | 38 (34 analyzed)                                   | 8                                                                |
| <b>Mean Age (SD)</b>          | 28.5 ( $\pm$ 8.6)                                                    | 22.9 ( $\pm$ 2.6)                                  | 34.5 ( $\pm$ 9.3)                                                |
| <b>Visual Status</b>          | Normal vision                                                        | Normal or corrected-to-normal vision               | Congenitally blind (7 completely, 1 with faint light perception) |
| <b>Eye Movement Recording</b> | Video eye tracking (SMI ETG 2 W, 120 Hz) + EOG (Smarter EEG, 500 Hz) | EOG (g.Hlamp, 512 Hz)                              | EOG (BioSemi EEG, 256 Hz)                                        |
| <b>Electrodes Setup</b>       | 6 electrodes (outer canthi + above/below left and right eye)         | 4 electrodes (outer canthi + above/below left eye) | 64-channel whole-head EEG, EOG channels used                     |
| <b>Procedure</b>              | 5 blocks alternating 190 s light/dark                                | 2 min eyes-closed during rest                      | 4 blocks of soundscapes (~15 min total)                          |
| <b>Lighting Condition</b>     | Alternating light/dark                                               | Eyes closed, dim room                              | Eyes closed, dim room                                            |
| <b>Additional Tasks</b>       | None                                                                 | None                                               | Soundscape identification (no response)                          |

## 2.8. Eye tracker

### 2.8.1. Noise and artifact rejection

Gaze and pupil data were first subject to visual inspection to detect noisy data. Following this, an automatic artifact rejection and noise reduction routine was applied to gaze and pupil data, separately, as described below.

**Gaze data:** For saccade detection, the gaze data from each eye (in total 4 gaze data streams: x direction and y direction, per left and right eye) was first linearly interpolated over missing values and then filtered with a 1 Hz high pass filter (Butterworth, zero-phase, filter order = 2). Then values exceeding 15 times the std from the mean as well as solitary values were replaced with NaNs. The time variation of the standard deviation was then calculated on the resulting time series using a moving window of 200 ms. Time periods in which the values exceeded a threshold of 100 were set to NaNs. Finally, NaNs were replaced by the cumulative sum and the sampling rate was increased based on the minimum distance of 2 consecutive samples in the four gaze data streams and then down sampled to 500 Hz to match each other. This step was important as the video samples from the two cameras did not perfectly match in time. After resampling, each sample would correspond to the same time point for each eye and additionally match the sampling rate of the EOG data.

**Pupil radius data:** The pupil data was visually inspected to exclude periods of low data quality. Such periods were mainly due to loss of pupil fit, which resulted in fast jumps between consecutive samples. Pupil data was subsequently equalized in sampling rate and up-sampled to 500 Hz using the same approach as for the gaze data. Otherwise, no additional technique for noise reduction was applied.

### 2.8.2. Blink detection using the eye tracker

We used a blink detection algorithm based on pupil radius as already described in (Brych & Händel, 2020). To obtain the blinks, we first calculated the z-value of the pupil radius data and then high pass filtered the data at 0.1 Hz (Butterworth, zero-phase, filter order = 2). A blink was then detected as time periods where the z-transformed radius from the left or right eye was below a threshold of -1 or missing for more than 3 consecutive samples. Blink start was defined by the last value above threshold, and blink end was defined by the last value still below the threshold. Blinks that were less than 4 samples apart from one another were concatenated. Blinks that were longer than 500 ms and shorter than 20 ms were discarded. Blink rate was calculated as blinks per minute. Only artifact free data was included in the blink rate calculation. A binocular blink was registered if a left eye blink and a right eye blink overlapped in time. Blink time was defined by the time of overlap.

### 2.8.3. Saccade detection using the eye tracker

Saccadic events were detected based on eye-movement velocity, a method developed by Engbert and Kliegl (2003). An elliptic threshold with a radius around the root mean square of the velocity values was applied. If the squared sum of the x and y velocity exceeded the threshold, a saccade was assumed. The threshold was set to 1 times the root mean square of the velocity values. The saccade velocity was calculated using 5 data points.

Please note that the following approach was identical to the one used for the EOG data. In a first step, only events with a minimum of 2 consecutive samples crossing the threshold were considered. Events were discarded if they occurred in opposite direction in close concession (<50 ms) unless the amplitude of one of the events was more than double in size compared to the other event. Saccadic events that were less than 15 ms apart were combined. Finally, we considered as saccades only those events that were longer than 8 ms and shorter than 200 ms. Saccade start was defined by the first value above half the threshold, and saccade end was defined by the last value still above half the threshold. Saccade rate was calculated as saccades per second. Only artifact free data was included in the saccade rate calculation. The saccade

amplitude was based on the Euclidian distance of the vertical and horizontal amplitude, using the maximum amplitude in the vertical and horizontal directions. The velocity calculation was based on the first derivative of a 5-sample time window, i.e. the mean of the first 2 samples minus the mean of the last 2 samples within this time window (Engbert & Kliegl, 2003).

A binocular saccade was assumed if a left eye saccade and a right eye saccade coincided (only relevant for experiment 1). Saccade time was defined by the time of overlap. Horizontal and vertical saccade amplitude was calculated as the mean of the maximum amplitude over both eyes within the binocular time window.

## 2.9. Main sequence

The main sequence (Bahill, Clark, & Stark, 1975) of the detected saccadic events was plotted as log(peak velocity) against log(amplitude). Please note that the saccade velocity was calculated following the approach by (Engbert & Kliegl, 2003) using 5 data points.

## 2.10. Saccade direction

Main saccade direction was calculated on those saccades that had an amplitude in the horizontal direction twice as large as in the vertical direction, or vice versa. These saccades were selected, and the percentage of directional saccades was calculated as the number of directional saccades divided by the number of all saccades times 100, separately for vertical and horizontal biased saccades. Following this, a directional index was calculated as the percentage of horizontal saccades divided by the percentage of vertical saccades. Accordingly, a value larger than 1 indicates that more saccades were executed with a directional bias in the horizontal direction, whereas a value smaller than 1 indicates that more saccades were executed with a directional bias in the vertical direction.

## 2.11. Statistical analysis

Before any statistical comparison was conducted, outliers and low-quality data were removed. To this end, we excluded datasets for which more than 80 % was subject to artifact rejection (as described above). For experiment 1 this meant that specific blocks of a subject's dataset were excluded if they had low quality. Data quality was separately assessed in the EOG data and in the eye tracking data and additionally, within the eyetracker data separately for pupil radius data and gaze data. Accordingly, a different number of blocks might have been included in the analyses for EOG and eye tracker data as well as for saccade and blink detection. For experiments 2 and 3, artifact rejection was done on the EOG data only and the whole dataset of subjects was excluded. Accordingly, the number of included datasets was identical for saccade and blink detection. For experiment 1, an additional outlier rejection on blocks and subject was applied using MAD (median rule, which uses the median absolute deviation from the median) with a threshold of 2 (Leys, Ley, Klein, Bernard, & Licata, 2013), i.e. on the distribution of combined data (subjects x blocks). A table of included datasets/blocks, as well as blink events and saccades (mean over participants +/- SE), is provided in supplementary Table S1. Please note that the absolute number of eye movement events can be different between methods while the rate is similar. This is because the noise free time can also differ between eye tracking gaze data, eye tracking pupil data and EOG data. Indeed, the data quality of EOG and eye tracking data is largely independent, as the noise in the EOG is mostly caused by muscle artifacts and external electrical noise, whereas the eye tracking data quality may vary due to, e.g., a change in the tracker position with respect to the eyes.

To compare the eye movement parameters between groups and conditions we used a Kruskal Wallis Test to compare blink rates, blink length, saccade rate and saccade amplitude and preferred directions, as

some of the data was not normally distributed. Post hoc comparisons were based on a Wilcoxon rank sum test.

To compare eye movement detection between EOG and eye tracking, we tested for correlations between the two data types. As we expect a strictly linear relationship between eye parameters referring to onset times we used a Pearson correlation. Please note that the data was not particularly skewed, and an outlier detection was applied.

## 2.12. Surrogate data

In order to investigate saccade dynamics in the experiments that did not include camera-based tracking, namely, when sighted participants sat with their eyes closed (experiment 3) and in a blind group of participants, we created a surrogate dataset and ran the saccade detection algorithm on it. To this end, the time series of each EOG electrode was shuffled so that each sample fell on a new random point in time. Only after this, were all other processing steps executed.

## 3. Results

### 3.1. Comparison between EOG and video-based eye tracking

In this first part we report on the comparison of several eye movement related parameters between two different eye tracking methods: EOG and video-based eye tracking. We want to point out that not all are expected to correlate at the same strength as the two different recording methods have some fundamental difference. Some parameters, such as eye movement rate, should be the same independent of the method and we indicate the assumed correlation with a correlation coefficient of 1 by plotting the diagonal. On the other hand, parameters like the duration of a blink, or the amplitude of a saccade, might only weakly or not at all correlate between detection methods. This is because for the video-based method blinks are detected based on the visual information from the pupil, while for the EOG the deflection of the electrical signals from around the eye are analyzed. Based on these theoretical considerations, we would not plot a diagonal for these parameters. The considerations will be discussed in detail below.

### 4. EOG and camera-based blink detection in light are correlated

We analyzed the blink patterns of sighted individuals who were freely moving their eyes in a lit room, as recorded using a video-based tracker and EOG electrodes (experiment 1, eyes open condition). The comparison of blink detection based on pupil size (video-based eye tracker) and vertical EOG deflection (Fig. 1) shows that the identified blink rates were correlated between detection methods. A Pearson correlation confirmed a significant positive correlation  $r(11) = [0.62]$ ,  $p = [0.031]$  (Fig. 1a). As the rate of blinks should be independent of the methods, we expected a correlation close to the diagonal. Although the blink rate is overall correlated, some subjects show a considerable difference in blink rate. We therefore tested if the blink rate calculation is based on the same blinks. On average, only 65 %  $\pm$  26(SD) of binocular EOG blinks are also detected in the binocular eye tracking data (Fig. 1b). If we take the monocular blinks of the eye tracker into account, we reach a higher percentage, 86 %  $\pm$  22(SD) of monocular EOG blinks that are also detected in the binocular eye tracking data (Fig. 1b). An average of 80 %  $\pm$  17(SD) of binocular eye tracking blinks are detected with binocular EOG (Fig. 1c). This suggests that the video-based eye tracker had a more variable detection performance over subjects. Accordingly, we compared the detected blink rate between the left and the right eye within one method. The blink rate correlated between eyes for the eye tracker measure  $r(11) = [0.88]$ ,  $p < [0.001]$  (Fig. 1e) and the EOG  $r(14) = [0.91]$ ,  $p < [0.0001]$  (Fig. 1f). Note that this correlation was much stronger for the EOG (most participants fell on the diagonal line) as opposed to the eye tracking method. Fig. 1f shows the difference between the expected correlation (diagonal) and the

actual difference between the left and right eye. EOG measurements of the subjects show a variance of 6, video eye tracking showed a variance of 30 with respect to the difference between expected and actual difference between the left and right eye. Accordingly, we assume that the similarity in eye movement parameters at least partly depends on the data quality. Data quality was rather variable over subjects and on average, 43 %  $\pm$  36(SD) of the eye tracker pupil data was included after artifact rejection and 88 %  $\pm$  117(SD) of EOG data was considered artifact free (supplementary Table S1).

As expected, blink length did not correlate between methods  $r(10) = [-0.14]$ ,  $p = [0.67]$  (Fig. 2a). Concerning blink latency, we found an average onset difference of 0.025 s, that is, EOG blinks were detected, on average, 25 ms earlier than eye tracker blinks. This timing difference was relatively consistent across participants (SD = 0.009 s), and likely stemmed from the distinct detection methods used. In eye tracker data, blink onsets are identified by the eyelid covering the pupil. In contrast, EOG detects blink onset via changes in the vertical electrooculographic (vEOG) signal, which are introduced by muscle activity around the eye and shifts in the cornea-retinal potential. These changes start before the eyelid has covered a significant portion of the pupil. Such methodological differences likely account for the timing differences in the detection of the blink onset (Fig. 2b).

### 5. Single subject

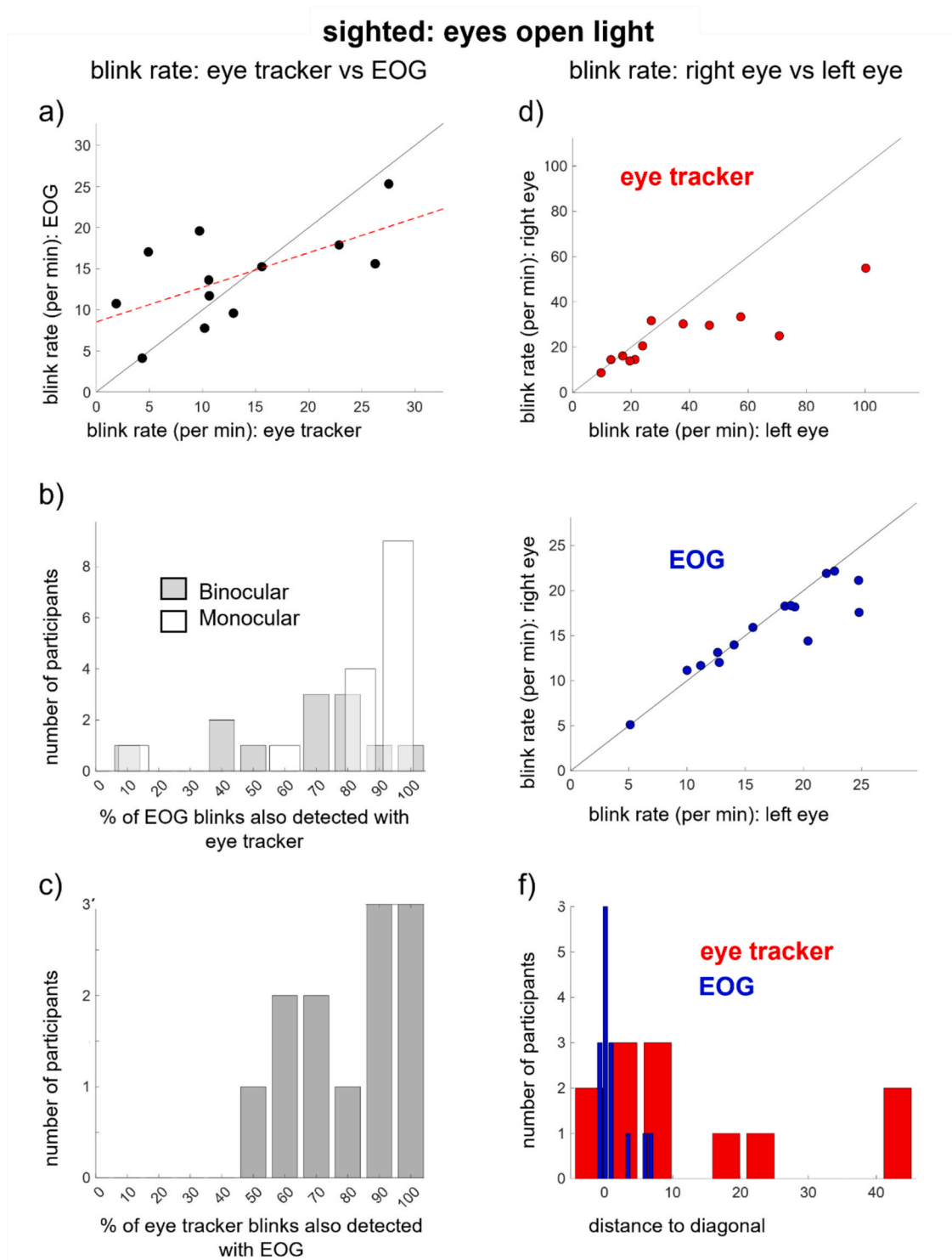
As the strength of correlation between methods likely depends on data quality, we inspect a participant that had 96 % artifact free eye tracker gaze data, 98 % artifact free eye tracker pupil data and 100 % artifact free EOG data. For these subjects we find 91 %  $\pm$  11 (SD) of binocular EOG blinks also detected in the binocular eye tracking data and 99 %  $\pm$  2 (SD) of binocular eye tracking blink also detected with EOG (SD is measured over the 5 trials). Looking at this subject separately (Fig. 3a), we find a positive linear relationship between blink rates calculated on the EOG data and those calculated on the eye tracker data. Since there were only 5 data points, no statistical evaluation was applied. All conclusions are based on group analysis.

### 6. EOG and camera-based saccade detection in light is correlated

We next compared saccade detection based on velocity measures in the eye tracking data to the EOG data. This comparison shows that the identified saccade rates are correlated, and indeed, a Pearson correlation indicated a significant positive correlation  $r(14) = [0.6]$ ,  $p = [0.018]$  (Fig. 4a). Find the saccade rate comparison over trials for a single subject in Fig. 3c. Please note that the EOG has a sampling rate of 500 Hz while the video-based eye tracker has a sampling rate of 120 Hz. The EOG is therefore more likely to pick up very short saccades. Also the saccade amplitude was significantly positively correlated  $r(14) = [0.64]$ ,  $p = [0.01]$  (Fig. 4b). The saccade amplitude correlation is shown for a single participant in Fig. 3d, and for all individual participants in the supplementary information (Fig. S1). Saccades further showed the expected amplitude velocity relationship (main sequence) in both methods (for subject 1 see Fig. 3b, for all individual subjects see Fig. S1). Saccade length was not significantly correlated between eye tracking methods  $r(14) = [0.32]$ ,  $p = [0.24]$  (Fig. 4c). We further plotted the velocity profiles separately for saccades in eye tracking data (red) and EOG (blue) (Fig. 4d). The profile is very comparable between methods, showing a steep increase in velocity at the onset of the saccade. This is expected as saccade detection was based on an increase in velocity.

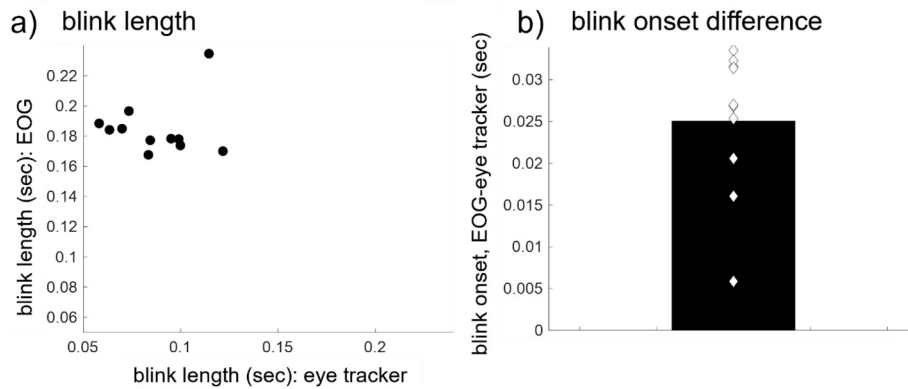
### 7. EOG and camera-based blink detection in darkness is correlated

We next turn to the comparison of blink detection based on pupil size (video-based eye tracker) and vertical EOG deflection when sighted



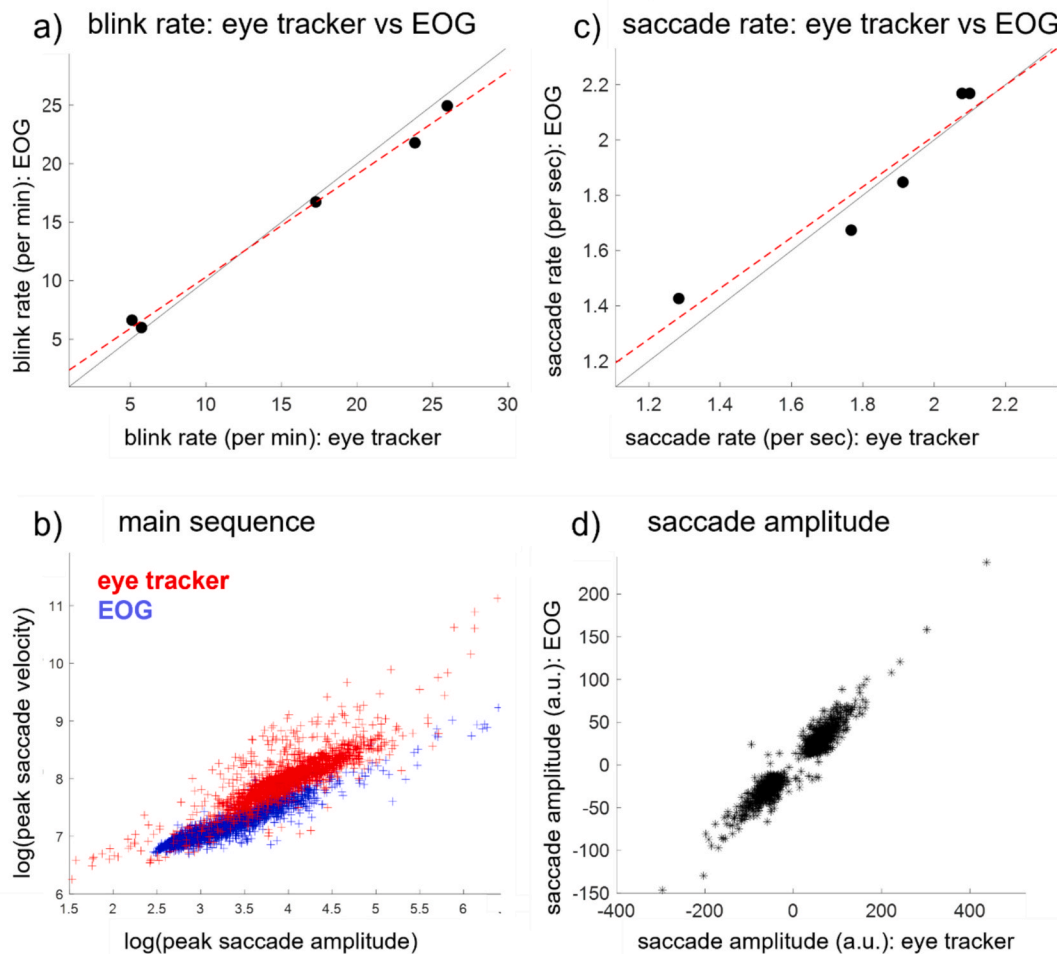
**Fig. 1.** Comparison of blinks detected with a video-based eye tracker and EOG in freely viewing sighted individuals in light. **a)** The blink rate calculated on the EOG data (y-axis) is plotted against the blink rate calculated on the eye tracker data (x-axis). Pearson correlation showed a significant positive correlation  $r(11) = [0.62]$ ,  $p = [0.031]$ . The regression line is plotted in red. Each data point represents one subject. The diagonal line marks the theoretical optimum, i.e. when both methods detect the same blinks. **b)** Histogram on the percentage of EOG blinks that were also detected with the eye tracker either when considering binocular or monocular eye tracking blinks. **c)** Histogram on the percentage of eye tracker blinks that were also detected with EOG. **d)** Same as (a) but now the blink rate obtained from the eye tracker data of the right eye (x-axis) is plotted against the left eye (y-axis). The blink rate from the eyes is positively correlated  $r(11) = [0.88]$ ,  $p < [0.001]$ . **e)** Same as (c) but now plotted for the right and left EOG eye data with a significant correlation  $r(14) = [0.91]$ ,  $p < [0.0001]$ .

## sighted: eyes open light

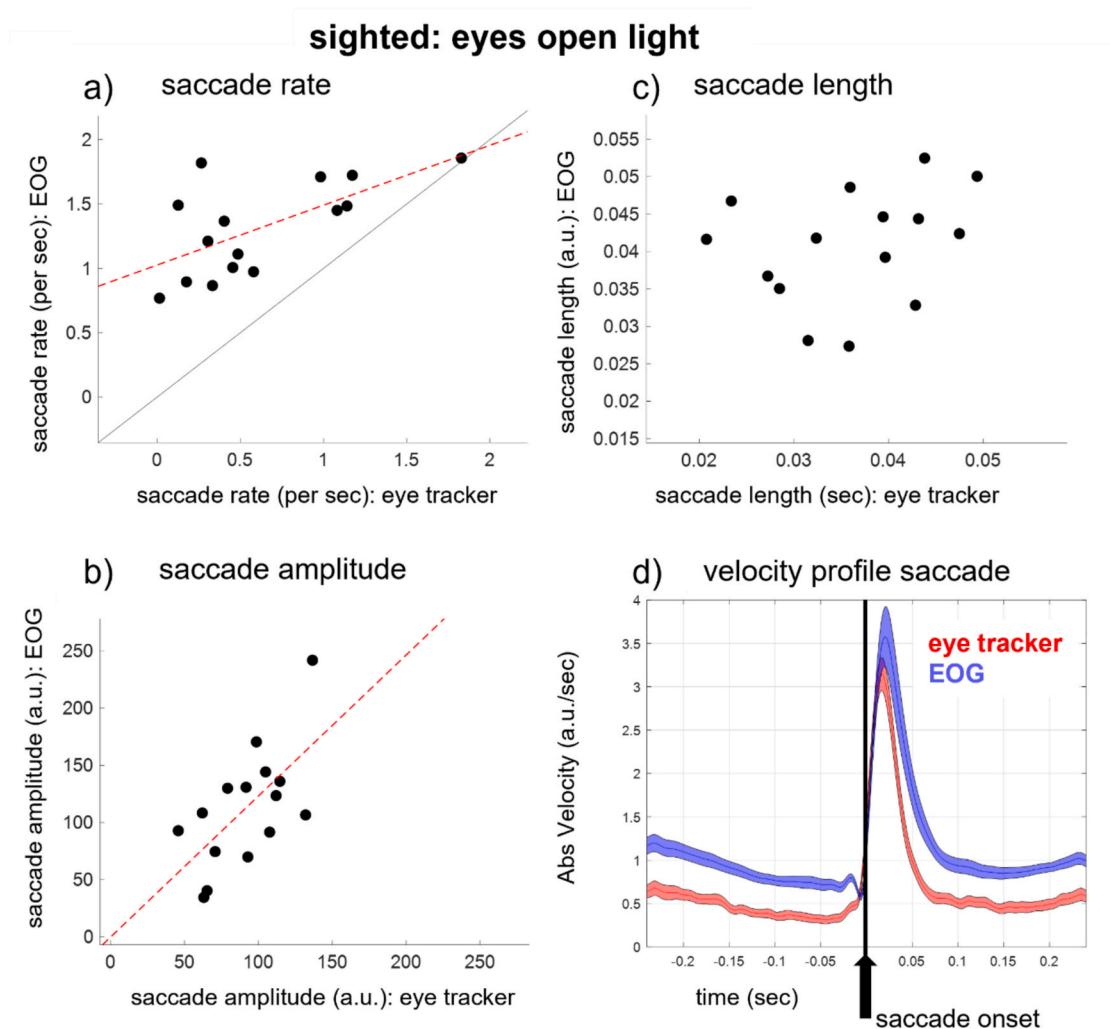


**Fig. 2.** Comparison of blink parameters detected with a video-based eye tracker and EOG in freely viewing sighted individuals in light. **a)** The blink length calculated on the EOG data (y-axis) is plotted against the blink length calculated on the eye tracker data (x-axis). There was no significant correlation  $r(10) = [-0.14]$ ,  $p = [0.67]$ . Each data point represents one subject. **b)** The latency of EOG-detected minus eye tracker-detected blinks in seconds. The average difference is 0.025 s.

## sighted single subject: eyes open light



**Fig. 3.** Comparison of blink rate and saccade properties detected with a video-based eye tracker and EOG in light in a **single subject** with low data noise. **a)** The blink rate calculated on the EOG data (y-axis) is plotted against the blink rate calculated on the eye tracker data (x-axis). Each data point represents one trial of 1 min length. The regression line is plotted in red. The diagonal line marks the theoretical optimum, i.e., when both methods detect the same blinks. **b)** Same as (a) but the saccade rate is plotted. **c)** The logarithm of the saccade velocity is plotted against the logarithm of the saccade amplitude. This representation is referred to as *main sequence*. A clear positive relationship is visible for saccades detected by EOG (blue values,  $N = 1764$ ) as well as by the eye tracker (red values,  $N = 1719$ ). **d)** The amplitude of saccades detected with EOG (y-axis) is plotted against the amplitude of the same saccades detected with the eye tracker (x-axis). Each dot marks a single saccade ( $N = 1061$ ). 5 % of residuals are more than 2 SD away from the regression.

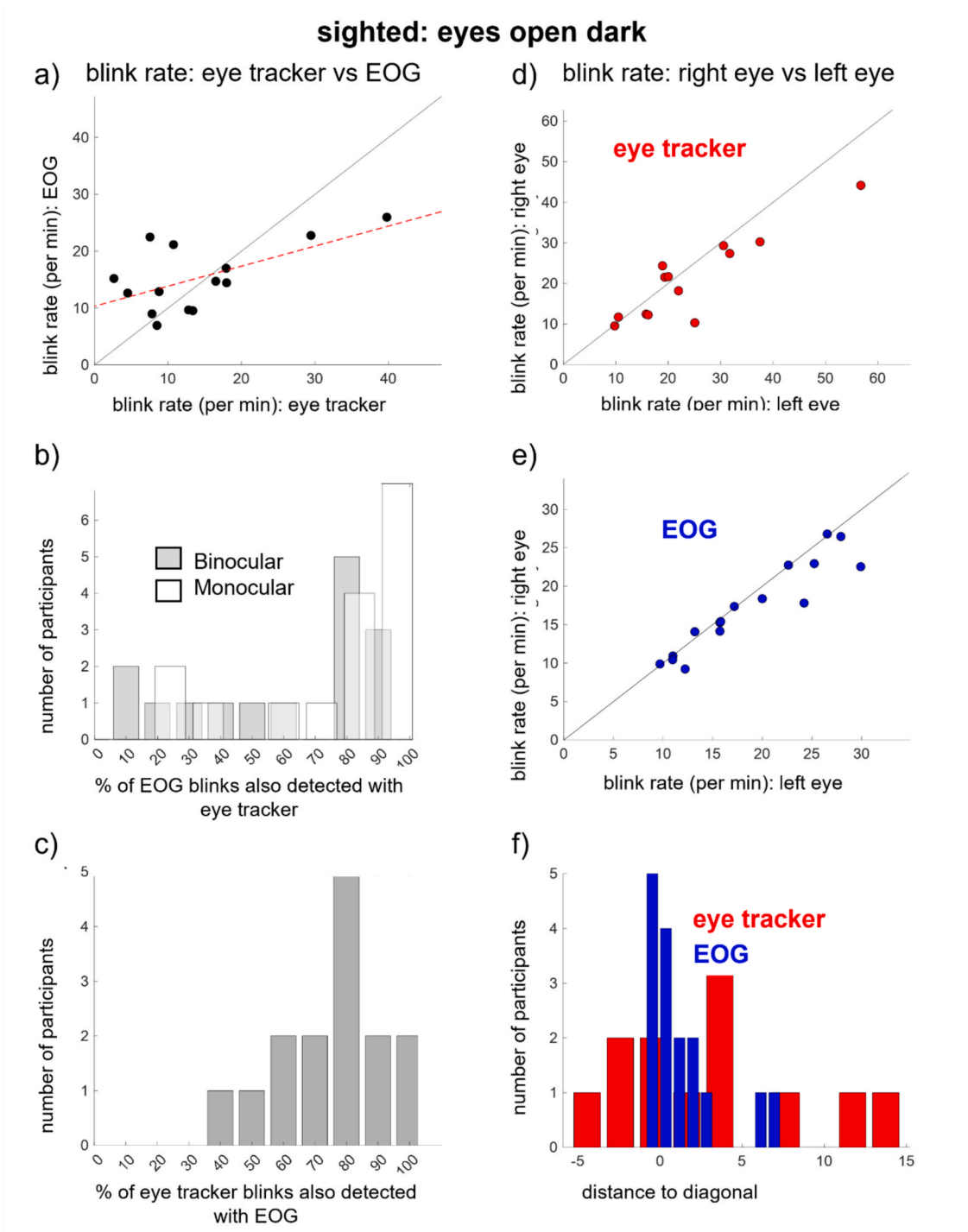


**Fig. 4.** Comparison of saccades detected with a video-based eye tracker and EOG in freely viewing sighted individuals in light. **a)** EOG saccade rate (saccades per second) is plotted against eye tracker saccade rate. The line marks the diagonal, the red broken line is the regression line. A Pearson correlation showed a significant positive correlation  $r(14) = [0.64]$ ,  $p = [0.01]$ . Each dot is a single participant. **b)** Same as (a) but the saccade amplitude (averaged over all saccades) is plotted. Saccade amplitude between EOG detected and eye tracker detected data was significantly positively correlated  $r(14) = [0.64]$ ,  $p = [0.01]$ . The regression line is plotted in red, however, the diagonal is missing because we do not expect the same amplitude value due to the difference in methods. **c)** Same as (a) but the saccade length (averaged over all saccades) is plotted. Saccade length between EOG detected and eye tracker detected data was not significantly correlated  $r(14) = [0.28]$ ,  $p = [0.31]$ . **d)** The absolute velocity profiles for saccades are plotted for eye tracking data (red) and EOG data (blue), separately. Time 0 marks the onset of the saccade, shaded area marks the standard error.

participants were sitting in complete darkness with their eyes open and free to move their eyes (experiment 1; condition dark). A Pearson correlation shows that the blink rates between the two methods have a significant positive relationship ( $r(13) = [0.6]$ ,  $p < [0.023]$ ) (Fig. 5a). Similar to the light condition, on average 60 %  $\pm$  30(SD) of binocular EOG blinks are also detected with the eye tracking data (Fig. 5b). If we take the monocular blinks of the eye tracker into account, we reach a increased percentage, 76 %  $\pm$  25(SD) of monocular EOG blinks that are also detected in the binocular eye tracking data (Fig. 5b). 74 %  $\pm$  17(SD) of binocular eye tracking blinks are detected with EOG (Fig. 5c). On average, 76 %  $\pm$  5(SD) of the eye tracker pupil data was included after artifact rejection and 93 %  $\pm$  8(SD) of EOG data was considered artifact free. Again, this suggests that the video-based eye tracker had a more variable detection performance over subjects. Comparing the detected blink rate between the left and the right eye within one method we find that the blink rate correlated between eyes for the eye tracker measure  $r(12) = [0.9]$ ,  $p < [0.0001]$  (Fig. 5d) and the EOG  $r(15) = [0.94]$ ,  $p < [0.0001]$  (Fig. 5e). The difference between the expected correlation (diagonal) and the actual difference between the left and

right eye is shown in Fig. 5f. EOG measurements of the subjects show a variance of 5.6, video eye tracking showed a variance of 33 with respect to the difference between expected and actual difference between the left and right eye. Data quality was less variable over subjects as compared to the light condition and on average, 77 %  $\pm$  14(SD) of the eye tracker pupil data was included after artifact rejection and 94 %  $\pm$  8(SD) of EOG data was considered artifact free (supplementary Table S1).

As in light, blink length did not correlate between methods  $r(10) = [-0.2]$ ,  $p = [0.49]$  (Fig. 6a). Concerning blink latency, we found an average onset difference of 0.025 s, that is, EOG blinks were detected, on average, 22 ms earlier than eye tracker blinks. This timing difference was relatively consistent across participants (SD = 0.0088 s), and likely stemmed from the distinct detection methods used. In eye tracker data, blink onsets are identified by the eyelid covering the pupil. In contrast, EOG detects blink onset via changes in the vertical electrooculographic (vEOG) signal, which are introduced by muscle activity around the eye and shifts in the cornea-retinal potential. These changes start before the eyelid has covered a significant portion of the pupil. Such



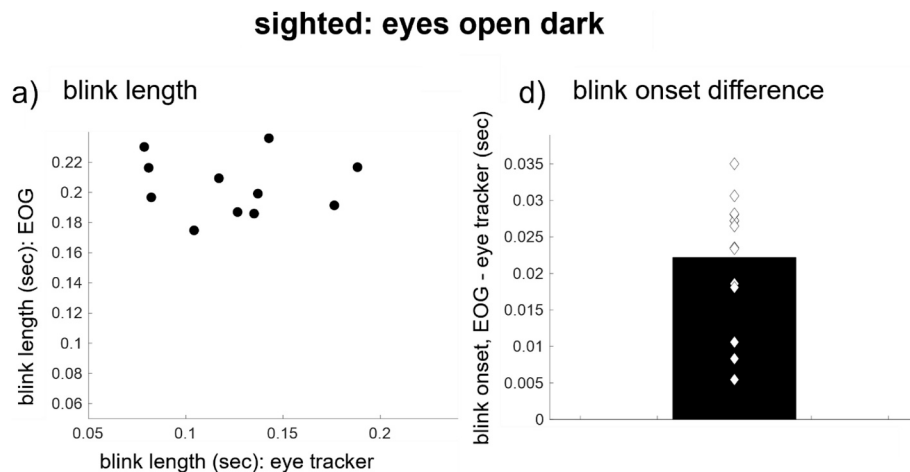
**Fig. 5.** Comparison of blinks detected with a video-based eye tracker and EOG in freely viewing sighted individuals in darkness. **a)** The blink rate calculated on the EOG data (y-axis) is plotted against the blink rate calculated on the eye tracker data (x-axis). Pearson correlation showed a significant positive correlation  $r(13) = [0.6]$ ,  $p = [0.023]$ . The regression line is plotted in red. Each data point represents one subject. The diagonal line marks the theoretical optimum, i.e. when both methods detect the same blinks. **b)** Histogram on the percentage of EOG blinks that were also detected with the eye tracker either when considering binocular or monocular eye tracking blinks. **c)** Histogram on the percentage of eye tracker blinks that were also detected with EOG. **d)** Same as (a) but now the blink rate obtained from the eye tracker data of the right eye (x-axis) is plotted against the left eye (y-axis). The blink rate from the eyes is positively correlated  $r(12) = [0.9]$ ,  $p < [0.0001]$ . **e)** Same as (d) but now plotted for the right and left EOG eye data with a significant correlation  $r(15) = [0.94]$ ,  $p < [0.0001]$ .

methodological differences likely account for the timing differences in the detection of the blink onset (Fig. 6b).

## 8. Single subject

If we inspect a participant that had over 98.2 % artifact free eye

tracker pupil data and 100 % artifact free EOG data, we find 56 %  $\pm$  14 (SD) of binocular EOG blink also detected in binocular eye tracker data and 95 %  $\pm$  5 (SD) of binocular eye tracker blinks also detected in EOG (SD is measured over the 5 trials). Looking at this participant separately (Fig. 7a), we find a positive linear relationship between blink rates calculated on the EOG data and those calculated on the eye tracker data.



**Fig. 6.** Comparison of blink parameters detected with a video-based eye tracker and EOG in freely viewing sighted individuals in darkness. **a)** The blink length calculated on the EOG data (y-axis) is plotted against the blink length calculated on the eye tracker data (x-axis). There was no significant correlation  $r(10) = [-0.2]$ ,  $p = [0.49]$ . Each data point represents one subject. **b)** The latency of EOG-detected minus eye tracker-detected blinks in seconds. The average difference is 0.022 s.

Since there were only 5 data points, no statistical evaluation was done.

### 9. EOG and camera-based saccade detection in darkness is correlated

We compared saccade detection based on velocity measures either in the eye tracking data or in the EOG data during darkness. The comparison shows that the identified saccade rates were correlated during darkness, similar to the case in light. A Pearson correlation confirmed a significant positive correlation  $r(15) = [0.64]$ ,  $p = [0.008]$  (Fig. 8a). Find the saccade rate comparison over trials for a single subject in Fig. 7c. Also the saccade amplitude was significantly positively correlated  $r(15) = [0.83]$ ,  $p < [0.0001]$  (Fig. 8b). The saccade amplitude correlation is shown for a single subject in Fig. 7d, and for all individual subjects in the [supplementary information \(Fig. S2\)](#) Saccades further showed the expected amplitude velocity relationship (main sequence) in both methods (for subject 1 see Fig. 8b, for all individual subjects see Fig. S2). Contrary to experiment 1, saccade length was significantly correlated between eye tracking methods  $r(15) = [0.56]$ ,  $p = [0.026]$  (Fig. 8c). We further plotted the velocity profiles separately for saccades for eye tracking data (red) and EOG (blue) (Fig. 8d). The profile is very comparable between methods, showing a steep increase in velocity at the onset of the saccade. This is expected, as saccade detection was based on an increase in velocity.

#### 9.1. Comparison between experimental groups: Sighted (light, dark, eyes closed) and congenitally blind

Blink rate and inter-blink interval do not significantly differ between sighted and blind.

After validating our EOG approach, we compared the outcome of blink and saccade detection in sighted participants in both light and darkness to sighted participants with their eyes closed, as well as congenitally blind individuals. Applying a Kruskal Wallis Test, we found no significant difference in blink rate ( $H(3,70) = 0.77$ ,  $p = 0.86$ ) (Fig. 9a) or inter-blink interval ( $H(3,69) = 0.51$ ,  $p = 0.92$ ) (Fig. 9b). However, there was a significant difference between experimental groups in blink length ( $H(3,70) = 13.67$ ,  $p = 0.0034$ ) (Fig. 9c). Post hoc comparisons based on a Wilcoxon rank sum test revealed a significant difference between all pairs except for two (sighted light vs. sighted eyes closed; and sighted dark vs. sighted eyes closed). Significant differences were found for sighted in light and two other groups (vs. sighted in darkness,  $Z(199) = -2.43$ ,  $p = 0.015$ ; vs. blind,  $Z(245) = 2.73$ ,  $p = 0.0064$ ). Significant differences were also found between the blind and

two other groups (vs. sighted in darkness,  $Z(250) = 3.03$ ,  $p = 0.0024$ ; vs. sighted with eyes closed,  $Z(821) = -2.87$ ,  $p = 0.0041$ ). A full table of post-hoc comparisons can be found in the supplementary [Table S2](#).

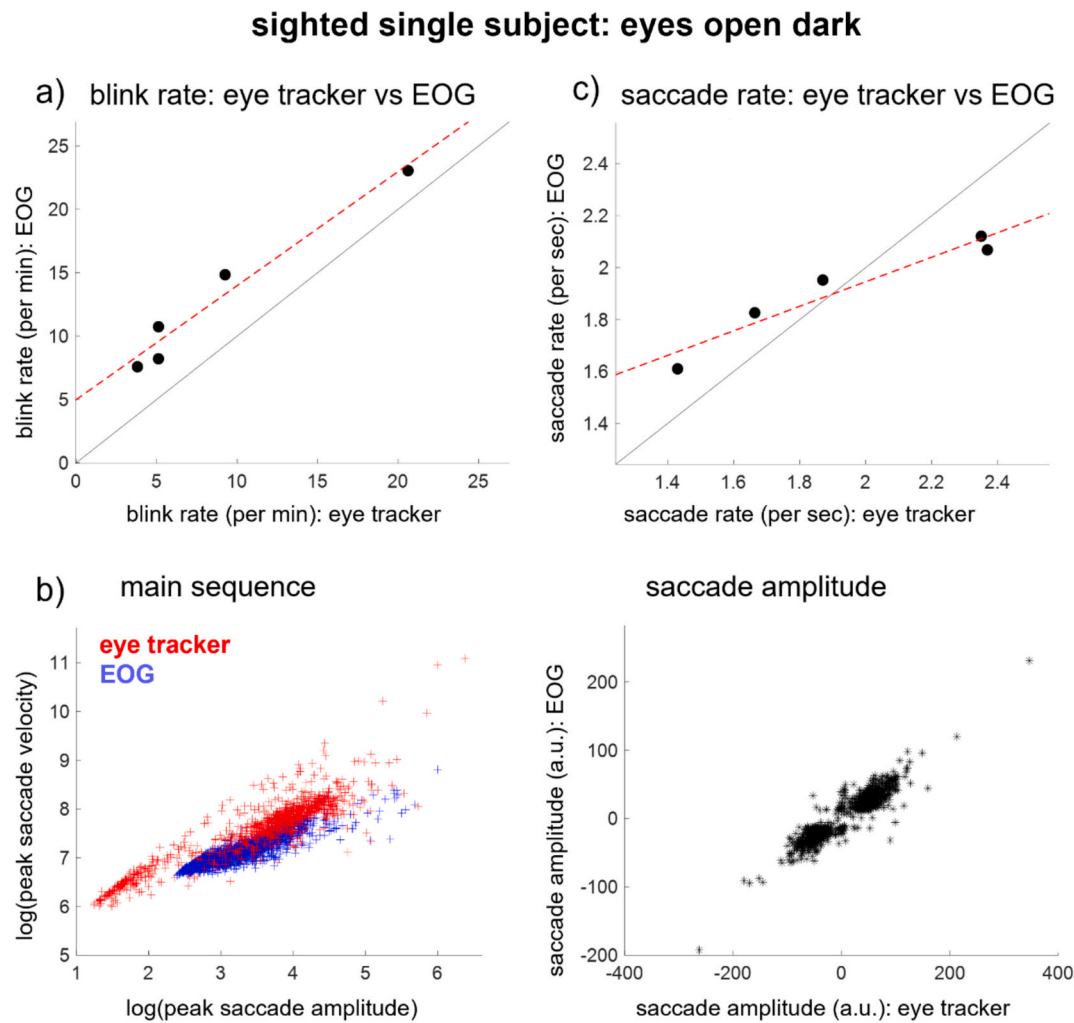
### 10. Saccade parameters significantly differ between sighted and blind

Comparing the outcome of saccade detection between the different experimental groups, we find a slightly different picture. Applying a Kruskal Wallis Test, we found an overall significant difference in saccade rate ( $H(3,70) = 22.11$ ,  $p < 0.0001$ ) (Fig. 9d). Post hoc comparisons based on a Wilcoxon rank sum test revealed a significant difference between sighted (in light or darkness) and sighted with eyes closed or blind (see [Table ST2](#)), but not between sighted with eyes closed and blind ( $Z(716) = -0.47$ ,  $p = 0.64$ ). There was no significant difference between sighted in light versus in darkness ( $Z(273) = 0.32$ ,  $p = 0.94$ ). Saccade amplitude ( $H(3,70) = 40.34$ ,  $p < 0.001$ ) (Fig. 9e) and the inter-saccadic interval ( $H(3,70) = 40.34$ ,  $p < 0.001$ ) (Fig. 9f) were significantly different between groups. For saccade amplitude and inter-saccadic interval post-hoc tests showed no significant difference between sighted in light and darkness ( $p > 0.6$ ), all other group comparisons showed a highly significant difference ( $p < 0.005$ ). The comparison of the sighted with eyes closed and blind showed a significant difference of  $p = 0.014$ . A full table of post-hoc comparisons can be found in [Table ST2](#) in the [supplementary material](#).

Interestingly, we also found an effect of preferred saccade direction ( $H(3,70) = 34.89$ ,  $p < 0.0001$ ) (Fig. 9g). Sighted individuals with eyes open, regardless of lighting condition, showed a preference for horizontal saccades. This preference for horizontal saccades in sighted individuals with eyes open is also significant in the eye tracking data (data not shown). Subjects with eyes closed or blind subjects showed a balanced rate of horizontal and vertical saccades. Post hoc comparisons based on a Wilcoxon rank sum test revealed no significant difference between sighted in light and darkness ( $Z(271) = 0.25$ ,  $p = 0.28$ ) and between sighted with eyes closed vs blind ( $Z(780) = 1.57$ ,  $p = 0.12$ ). All other differences were significant ( $p < 0.001$ ) (see [Table ST2](#)).

Finally, the slope of the main sequence was significantly different between all experimental groups (Kruskal Wallis:  $H(3,64) = 34.12$ ,  $p < 0.0001$ ) see [Table ST2](#)) except between sighted in light and blind (Post hoc Wilcoxon rank sum:  $Z(180) = 1.26$ ,  $p = 0.21$ ) (Fig. 9h). The main sequence, as observed when plotting the logarithm of the peak amplitude against the logarithm of the peak velocity of all individual saccades for each experimental group separately, can be observed in [Fig. S7](#).

Plotting the velocity profile of saccades separately for each



**Fig. 7.** Comparison of blink rate and saccade properties detected with a video-based eye tracker and EOG in darkness in a **single subject** with low data noise. The black line marks the diagonal and the broken red line the regression line. **a)** The blink rate calculated on the EOG data (y-axis) is plotted against the blink rate calculated on the eye tracker data (x-axis). Each data point represents one trial of 1 min length. **b)** Same as (a) but the saccade rate is plotted. **c)** The logarithm of the saccade velocity is plotted against the logarithm of the saccade amplitude. This representation is referred to as *main sequence*. A clear positive relationship is visible for saccades detected by EOG (blue values,  $N = 1820$ ) as well as by the eye tracker (red values,  $N = 1825$ ). **d)** The amplitude of saccades detected with EOG (y-axis) is plotted against the amplitude of the same saccades detected with the eye tracker (x-axis). Each dot marks a single saccade ( $N = 872$ ). Over 95 % of residuals are within 2 SD from the regression.

experimental group shows that the modulation of velocity is similar in all groups (Fig. 10a). When plotting the vEOG modulation around a detected blink (Fig. 10b), a larger amplitude can be seen for sighted with their eyes open (light and dark) compared to sighted with eyes closed and blind participants. This decreased deflection around a blink is likely the cause for the observed difference in blink length (Fig. 9c) between these participant groups.

## 11. Discussion

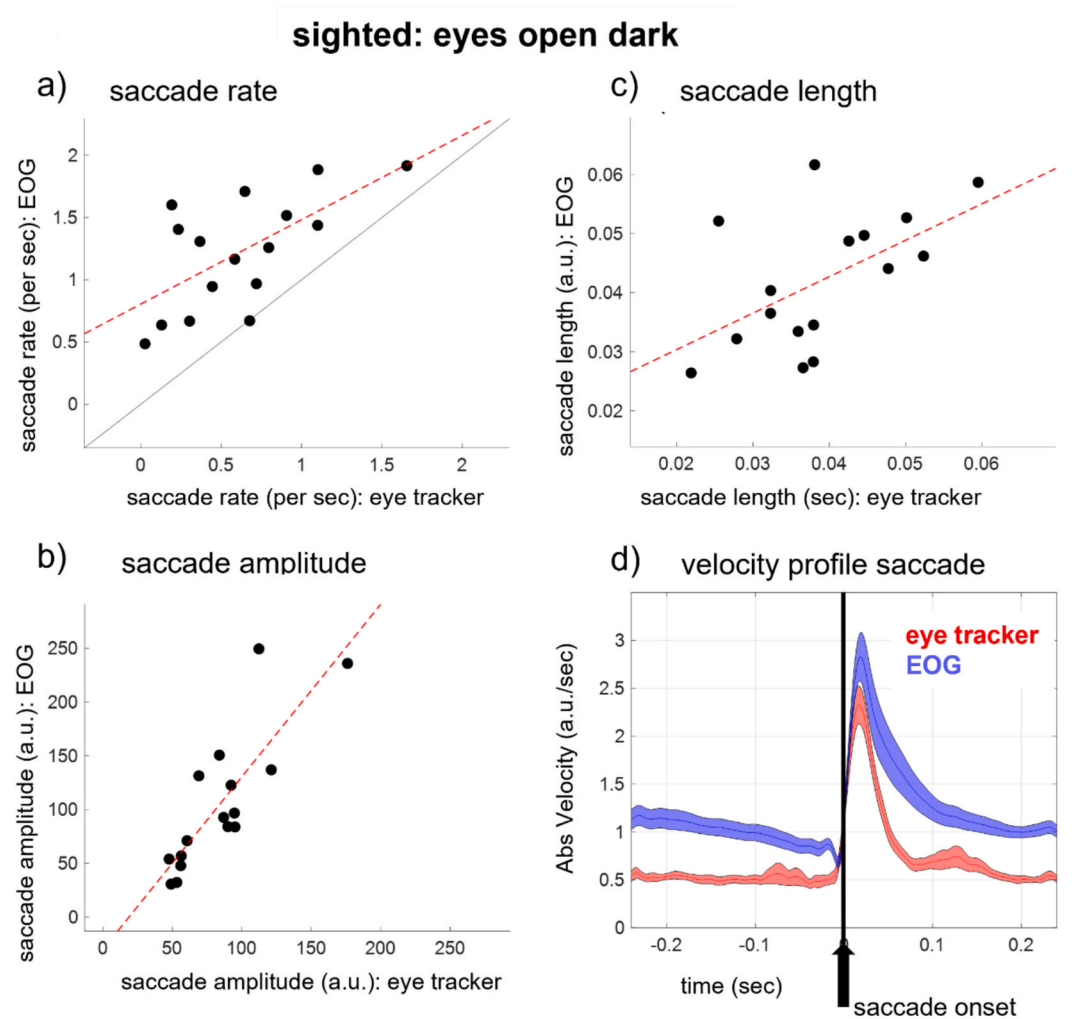
We quantified the ability of EOG to identify eye movements in a natural and largely unrestricted setup. Our study shows that blink rate, saccade rate, amplitude and direction can be detected with relatively high precision using EOG in comparison to video-based eye tracking. This is beneficial for setups in which the eyes are closed or under reduced control, such as in congenital blindness. In such setups, EOG could be used to detect these metrics with good precision, allowing for valuable comparisons of eye movement patterns between different eye states and vision conditions. In addition to validating the use of EOG, we

examined the eye-related movements of four groups of participants (sighted with eyes open in light and dark conditions, sighted with eyes closed, and blind). We demonstrate that the eye-related movements of sighted participants with eyes closed are comparable to those found in congenitally blind individuals.

Detection algorithm of EOG compared to video-based eye tracking.

As a first step, we aimed to validate our detection algorithms by comparing the detection quality of EOG-based detection to that of a video-based eye tracker. Since we were primarily interested in setups with little restrictions on eye and head movements, we chose an SMI 120 Hz video-based eye tracker (i.e., we excluded the use of a high-precision eye tracker, which requires a fixed head position).

Specifically, we examined comparisons between EOG and video-based eye tracker measurements of eye-movement related parameters in two groups of participants: sighted individuals in light, freely viewing conditions, as well as sighted individuals in darkness, freely viewing conditions. Overall, we showed a high correlation between EOG and eye tracker measurements for most eye-movement related parameters. Concerning saccades, in both light and darkness, saccade rate showed a



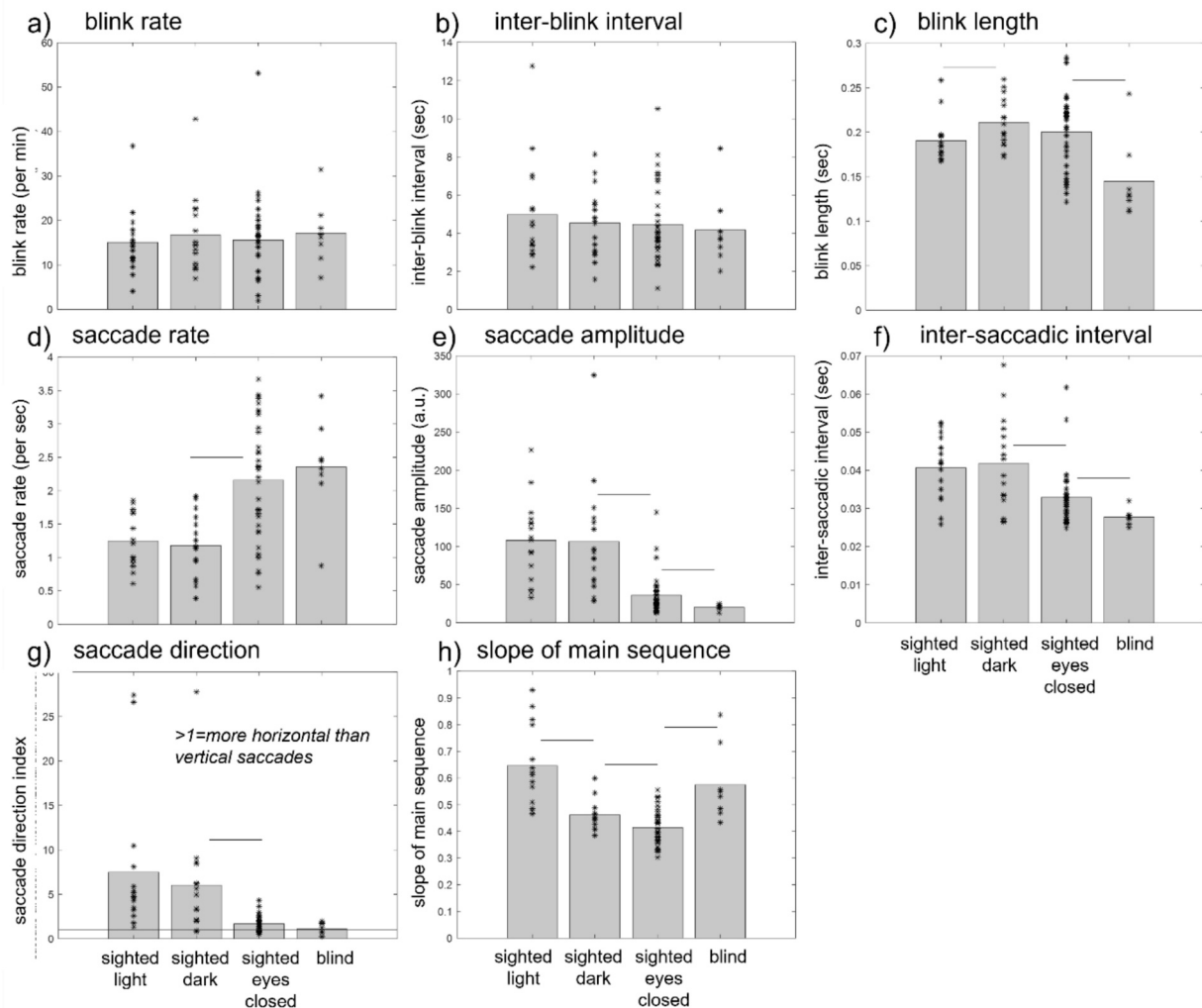
**Fig. 8.** Comparison of saccades detected with a video-based eye tracker and EOG in freely viewing sighted individuals in darkness. **a)** EOG saccade rate (saccades per second) is plotted against eye tracker saccade rate. The line marks the diagonal. A Pearson correlation showed a significant positive correlation  $r(15) = [0.64]$ ,  $p = [0.008]$ . Each dot is a single participant. **b)** Same as (a) but the saccade amplitude (averaged over all saccades) is plotted. Saccade amplitude between EOG detected and eye tracker detected data was significantly positively correlated  $r(15) = [0.83]$ ,  $p < [0.0001]$ . **c)** Same as (a) but the saccade length (averaged over all saccades) is plotted. Saccade length between EOG detected and eye tracker detected data was significantly correlated  $r(15) = [0.56]$ ,  $p = [0.026]$ . **d)** The absolute velocity profiles for saccades are plotted for eye tracking data (red) and EOG data (blue), separately. Time 0 marks the onset of the saccade, shaded area marks the standard error.

significant positive relationship between detection methods (see Fig. 4a and 8a). Note that this relationship was close to the identity line, meaning that the values were not only positively correlated but actually similar, however with a certain offset; in both conditions, saccade rate is higher for EOG detection. This can be likely explained by the difference in methods: the EOG has a sampling rate of 500 Hz while the video-based eye tracker has a sampling rate of 120 Hz. The EOG is therefore able to pick up shorter saccades. Saccade amplitude measurements in the two methods were not only correlated at the group level (see Fig. 4b and 8b), but showed a very clear positive relationship for each individual (see Fig. S1 and S2). This fits well to the findings that the velocity profiles around detected saccades were overlapping (see Fig. 4d and 8d). Saccade length only correlated during darkness (see Fig. 8c). As the video-based eye tracker estimates saccade amplitude and length based on the distance between previous and following fixation point but the EOG estimates amplitude based on the strength of the electrical signal of the eye muscles and the shift of the cornea-retinal potential, a correlation is expected, however, certain aspects need to be considered. While we can assume a linear relationship between the amplitude and length of the change in the cornea-retinal potential and the length and amplitude

of the saccade measured by the movement of the pupil, we cannot simply assume such a relationship between the strength and duration of the muscular command as a saccade as it is a ballistic eye movement. This might explain the less stable correlation between methods. We further found a preference for horizontal compared to vertical saccades (see Fig. 9g).

Importantly, the parameter values we present for the identification of saccades in both EOG and eye tracking data are consistent with previous reports in the literature. The saccade rate has been reported with variable frequencies, ranging from low frequencies of approximately 0.5 Hz (Steinman, Cunitz, Timberlake, & Herman, 1967) to higher frequencies of 3–5 Hz (Nuthmann, Smith, Engbert, & Henderson, 2010; Otero-Millan, Troncoso, Macknik, Serrano-Pedraza, & Martinez-Conde, 2008). Our average saccade rate of around 1.2 Hz for sighted individuals with eyes open is within the range of reported frequencies, particularly as it is known that saccade rate is highly influenced by the sensory input, task and instructions (Rolfs, 2009). Additionally, we are able to observe (see Fig. 3b and 7b and S1 and S2) the well described main sequence describing the linear relationship between saccade amplitude and velocity (Bahill et al., 1975). Given this body of work, our

## comparison of saccade and blink parameters between experimental groups

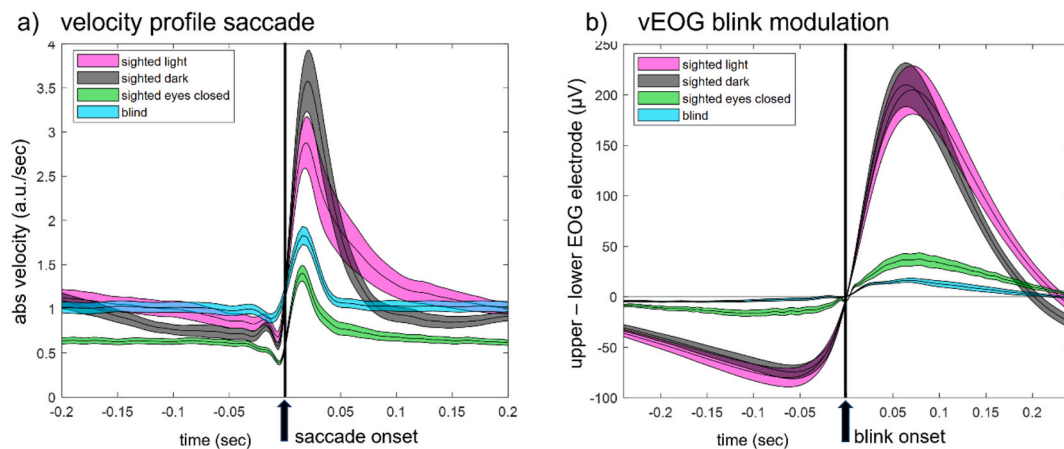


**Fig. 9.** The parameters concerning blinks and saccades are plotted and statistically compared between the experimental groups. ‘Sighted light’ and ‘sighted dark’ are sighted participants freely viewing with their eyes open in light or in complete darkness (experiments 1). ‘Eyes closed’ refers to sighted participants in the freely viewing setup, but with their eyes closed. ‘Blind’ participants are congenitally blind individuals. Parameters are **a)** blink rate, **b)** inter-blink interval, **c)** blink length, **d)** saccade rate, **e)** saccade amplitude, **f)** inter-saccadic interval **g)** saccade direction, **h)** slope of the main sequence. The bars mark the mean value, and the stars depict the individuals. All parameters that show a significant difference between groups, as by a Kruskal Wallis Test, were subject to a post-hoc test (based on Wilcoxon rank sum test). To preserve readability, only the significant ( $p < 0.05$ ) post-hoc comparisons between neighboring groups are indicated by a horizontal line. A full table of post-hoc comparisons can be found in **Table ST2** in the supplementary material.

identification results appear to be plausible.

Concerning blinks, we found that in both sighted groups (eyes open, light; eyes open, darkness) the blink rate correlates between EOG and eye tracking detection (see Fig. 1a and 5a). The correlation was close to the diagonal, confirming that the rate of blinks is largely independent of the detection method. Plausibility checks show a good correspondence between this study’s and previously reported blink rates. Spontaneous blinks have been reported to occur approximately 10 to 15 times a minute (Burr, 2005; Doughty, 2001; Kaminer et al., 2011). Our detected blink rate lies on average around 10 to 20 blinks per minute for both the eye tracking and the EOG method, corresponding well to the literature. However, some subjects showed a considerable difference in blink rate between methods. We propose that this is at least partly related to the individual data quality as data quality was rather variable over subjects and on average lower for the eye tracker pupil data (light, 43 %  $\pm$  36 (SD), dark: 77 %  $\pm$  14(SD)) than the EOG data (light: 88 %  $\pm$  117 (SD), dark: 94 %  $\pm$  8(SD); supplementary Table S1). Fig. 1 b/c and 5b/c also showed that EOG is more likely to pick up an eye tracker blink

than the other way around. This can be interpreted that either EOG is more reliable overall or that there are specific subjects where the signal quality of the SMI is worse than of the EOG. As for the blink rate we take only binocular blinks into account (i.e. blinks that are found in the left and right eye at the same time) the problem could be based on single eye data. Indeed, when we test for reliability between eyes, (Fig. 1 d/e, 5 d/e) EOG data from the left and right eye are correlated in a way that the distance to the diagonal is smaller by a factor of 5 compared to the eye tracker. Accordingly, if we take monocular data from the eye tracker (Fig. 1 b, 5b) we find almost a similar percentage of EOG blinks also detected by monocular eye tracking blinks. Also, when testing an individual with good data quality in both methods (>98 % good data), we find a very good correlation between methods for blink and saccade parameters (Figs. 3 and 8). Our findings suggest that the video-based eye tracker had, in our case, a more variable detection performance over subjects and that as soon as the signal quality is deteriorated, the detection overlap breaks down. This has also been observed when comparing mobile EOG and eye tracking in stationary vs. walking



**Fig. 10.** The saccade induced velocity changes, and the blink induced vEOG modulation per experimental group. Groups as described for Fig. 7. **a)** The velocity profiles for saccades are plotted for sighted in light (magenta), sighted in darkness (grey), sighted with eyes closed (green) and blind (cyan). The velocity values are plotted as absolute values as the saccades have different directions. Time 0 marks the onset of the saccade. The shaded area marks the standard error. **b)** Data from the electrode above the eye was subtracted from the data of the electrode from below the eye to create a vertical EOG signal (vEOG). This vEOG is plotted for  $\pm 250$  ms around, separately for each experimental group. The shaded area marks the standard deviation over subjects.

participants (Stuart et al., 2016).

When analyzing the blink length we find no correlation between techniques. While the length of eye tracker blinks vary over participants, measuring between 0.07 and 0.19 sec, EOG blinks have a rather stable length around 0.2 sec. Additionally, we find that there is a consistent difference in blink onset, with EOG-detected blinks occurring approximately 25 and 22 ms (during light and darkness) earlier than eye tracker-detected blinks. A feasible explanation for the difference again relates to the different signals used for blink detection. EOG blink detection is mainly based on the modulation of the vEOG, which is caused by muscle activation and the change in the cornea-retinal (or standing) potential during a blink (Matsuo, Peters, & Reilly, 1975). This is a rather standardized modulation once a blink is initialized. Our eye tracking blink detection, however, uses pupil size, which is reduced and then zero (as the pupil is not visible anymore) during a blink. Pupil size is only affected by a blink once the lid starts covering the pupil, and thus happens slightly later than the start of the blink.

In summary and in line with previous reports (Müller, Wendt, Kollmeier, & Brand, 2016; Picot, Charbonnier, Caplier, & Vu, 2012; Pleshkov et al., 2022; Vidal, Bulling, & Gellersen, 2011), we found strong evidence that EOG in a free viewing condition can be used to detect and quantify blinks and saccades, in a comparable fashion to a mobile video-based eye tracker with a 120 Hz sampling rate, and in a setup which makes no restrictions as to the execution of eye or head movements.

## 12. EOG detected eye movements under different conditions and experimental groups

### 12.1. Eyes open compared to eyes closed in sighted

In a second step, we analyzed EOG data in a setup in which a video-based eye tracker cannot be used. This is the case if the eyes are not visible or hardly visible, such as when the eyes are closed or in participants that have reduced control over their eye movements and eye lids (e.g., congenitally blind individuals).

We found that our algorithm still reliably detected blinks and saccades, thereby suggesting that eye closure does not prevent saccadic eye movements and blinks. Note that, when the eyes are closed, a blink is not indicated by the closure of the eye lid, but rather by muscle activity that leads to a sort of slight squinting. While there have been previous studies on saccades in individuals with their eyes closed, the focus was on movement speed or related brain activity, and it was always with respect to the instructed execution of large saccades (Hüfner et al., 2008; Shaikh

et al., 2010). This is fundamentally different to our no-task setup, which rather examined spontaneous eye movements. The detection of eye movements using EOG during eyes closed conditions and without instructed saccades is actually a frequently applied method in sleep stage classification. The rapid eye movement (REM) stage, characterized by relatively large saccades (visible at sight) at a rate of around 15 per minute (Takahashi & Atsumi, 1997), can be well detected with a single EOG electrode pair (Cooray et al., 2021). Whether small saccades are executed during sleep is still debated (Coakley, Williams, & Morris, 1979). Our findings indicate that during an awake state, blinks and saccades are present when the eyes are closed, but there are significant differences to the eyes open state with respect to rate, length, amplitude and other features, as will be discussed below.

### 12.2. Sighted vs blind participants

When we compared blinks in sighted individuals (under different conditions) and blind individuals, neither the rate nor the inter-blink interval was significantly different in any comparison (Fig. 9a, b) i.e. the blink rate and the inter-blink interval was similar in sighted in light, sighted in darkness, sighted with eyes closed and in blind. Indeed, blink rates have been reported to be normal in congenitally blind children (Hall, 1945) and adults (Ponder & Kennedy, 1927). The detected blink length, however, did show a significant difference, with blind subjects showing shorter blinks than sighted independent of the condition (see Fig. 9c, supplementary Table S2). Importantly, the shape of the blink-induced modulation of the vEOG was overall comparable but showed a reduced amplitude for sighted with eyes closed and blind participants (see Fig. 10b). As the modulation of the vEOG during a blink is largely based on the cornea-retinal (or standing) potential (Matsuo et al., 1975), an amplitude reduction during darkness is expected. We, however, did not observe amplitude differences between the light and darkness conditions. This was possibly due to our experimental design, which induced constant changes in lighting at a one-minute rate, i.e. the dark phase lasted one minute. Since a reduction in the cornea-retinal potential starts approximately 3 min after a change in light and reaches its maximum values after around 7–15 min (Constable et al., 2017), our experimental design during eyes open likely has prevented the retinal potentials from ever reaching full reduction so no significant difference in amplitude between light and dark conditions was found. Contrarily, participants with closed eyes as well as congenitally blind participants could be said to have experienced longer times of darkness potentially leading to a more visible reduction in the potential. Sighted participants

with eyes closed sat in a dimly lit room for preparation and then experienced darkness for 2 min due to eye closure. Blind participants are continuously in darkness and thus the lowest potential could be expected for this group, as observed in our data sets. However, it is also possible that the potential is reduced in parallel with the cause of blindness.

Unfortunately, we are not aware of information on the change in the cornea-retinal potential due to, e.g. retinopathy of prematurity, which was the prevalent cause for blindness in our participant pool. An indication of a change of the cornea-retinal potential provides work showing a reduced retinal potential measured by flash electroretinography in a rat model for retinopathy of prematurity (Hatzopoulos, Vessey, Wilkinson-Berka, & Fletcher, 2014). Interesting earlier work also nicely delineates the different influence of an intact eye and the movement of the eye lid on the blink-induced change in EOG potential (Matsuo et al., 1975). Comparing a healthy control with one participant with a glass eye on one side and one participant with a one-sided facial paralysis, they conclude that the generation of the blink-induced potential is introduced by the movement of the eyelid over an intact eye. As the eyes normally do not move strongly during a spontaneous blink, the change in potential is introduced by the eyelid moving over the cornea. Please keep in mind that the pronounced upward movement of the eyeball (Bell's Phenomenon) only occurs during a voluntary blink but not during involuntary blinks as confirmed with the search coil technique (Iwasaki et al., 2005). However, this earlier work by Matsuo and colleagues showed additional points. First, they showed that the positive and negative change in the upper and lower electrode is still visible but greatly reduced during a blink without eyelid movement by testing a subject with facial paralysis. Second, the study showed that the blink in a subject wearing one glass eye was still visible in the upper and lower electrode around the glass eye with a change in potential approximately between 60 and 90  $\mu$ V. The side with the functioning eye showed the classical pattern of a deflection in the upper electrodes, of around 400–500  $\mu$ V and an increase of about 150  $\mu$ V in the lower electrode. This suggests that blinks in blind participants with glass eyes or no eyes lead to a smaller change in voltage, possibly explaining the reduced vEOG amplitude. Unfortunately, we cannot retrace how many subjects in our participant pool had glass eyes or no eyes. The reduced blink-induced vEOG deflection, independent of its cause, likely underlies the observed shorter blinks in blind.

Concerning saccades, we observed the most pronounced difference between sighted individuals with eyes open (light and dark) and sighted individuals with eyes closed or blind participants. Concerning saccade rate, amplitude, inter-saccadic interval and preferred saccade direction (Fig. 10 d-g), data seems to mostly fall into two parameter ranges: sighted participants with their eyes one in one group and sighted with eyes closed and blind in the other group. Specifically, the saccade rate was significantly lower in the sighted eyes open participants compared to the other two groups of (sighted with eyes closed and blind), whereas the inter-saccadic interval and the saccade amplitude were significantly larger. That the inter-saccadic interval and the saccade rate are inversely related is expected. Also, the increase in saccade amplitude does make sense considering that visual input from the whole visual field might introduce large saccades in order to shift the gaze over the whole visual field. A higher number of large saccades might lead to a lower saccade number overall. On the other hand, it has been suggested that separate populations of saccades exist. Using an onset delay paradigm, it was shown that only a subpopulation of saccades was affected by a visual change (Henderson & Pierce, 2008; Henderson & Smith, 2009). Similarly, (Batten, 2018) reported one subpopulation of inter saccadic intervals (or fixation durations) that was modulated by auditory sensory changes, whereas another subpopulation showed a rather invariable and shorter duration of fixation (245–250 ms), which indicates a higher saccade rate. If sighted people have their eyes open, the system might mainly tap into a type of saccade that favors a lower saccade rate (i.e., longer periods of gaze). While eyes closed or blind, saccades that are

sensory independent and occur at a higher frequency might be prevalent. Another example of such a prevalence might be downbeat nystagmus as e.g. found in cerebellar patients (Hufner et al., 2007; Wagner, Glaser, Brandt, & Strupp, 2008) might further speak for a basic, sensory input-independent rhythm. Congenitally blind individuals have been described to exhibit spontaneous rapid eye movements, such as nystagmus, which have been demonstrated using both electrooculography (EOG) (Kömpf & Piper, 1987) and the magnetic search coil technique (Leigh & Zee, 1980; Schneider et al., 2013). The presence of nystagmus suggests that the saccade-generating mechanism remains at least partially intact. This notion is further supported by findings showing that, after sight restoration in individuals with long-standing congenital cataracts, visually guided gaze shifts could be performed as effectively as those of normally sighted individuals with nystagmus (Zerr et al., 2020).

Interestingly, the main direction for the saccades was also changed in our experiment. While saccade direction in sighted with eyes open is mainly executed in the horizontal direction, the direction of execution for sighted with eyes closed and blind is equally likely in the vertical as in the horizontal direction (see Fig. 10 g). This, again, is in line with the mentioned downbeat nystagmus as a basic saccade rhythm that might affect saccade direction and, in this case, counteracting the preference for horizontal saccades.

Importantly, the general features of saccades are met. Specifically, the main sequence of saccades which refers to the linear relationship between the log of saccade velocity and the log of saccade amplitude, is clearly seen in all measured groups (supplementary Fig S7). The slope, however, is significantly different between all groups except between blind and sighted (supplementary Table S2, Fig. 10 h). Can the reduction in slope be explained by the change in saccade direction? It is well documented that horizontal saccades are generally faster than vertical and oblique ones, due to a higher performance of the horizontal recti muscles (Vergilino-Perez et al., 2012; Gibaldi et al., 2016). However, while there is a significant reduction in horizontal saccades for blind compared to sighted with eyes open in light, there is no significant difference in the slope of the main sequence between these two groups. Accordingly, the underlying cause is likely a different one.

### 12.3. Comparison to saccade detection in surrogate data

Even though saccades in the sighted with eyes closed and in blind participants follow the main sequence, they showed a significant difference to in preferred direction, rate and amplitude compared to sighted with eyes open. Accordingly, we wanted to secure that these detected saccades in blind and sighted with eyes closed are genuine ones. To this end we ran our saccade detection algorithm on surrogate data with similar features by shuffling the original data. The saccade detection algorithm indeed also detected saccades in the shuffled data; however, in all experimental groups, the real saccades had significantly different characteristics in comparison to the surrogate data: saccade length was higher, saccade amplitude and rate was lower (except in two cases), and there was a shift of the preferred direction towards vertical (See supplementary material Figs. S3 – S8). To summarize, for all experimental groups, there were significant differences between the saccades detected in the shuffled data vs. saccades detected in the real data. This comparison supports the conclusion that saccadic eye movements are present in sighted with eyes closed and blind. Overall, we find that EOG is well able to detect eye related movements during eyes closed and in blind.

Why do sighted with eyes closed and blind have eye movements.

In a last step, we want to discuss why eye movements might still be prevalent in sighted individuals with eyes closed and in congenitally blind individuals even though no visual input will be processed. That eye movements such as blinks and saccades are present in these conditions could indicate that the role of those movements goes beyond sampling of visual information. The fact that eye movements are prevalent in blind

further suggest that eye related motor behavior is also not learned due to visual experience.

That eye movements are somehow independent of visual input has been indicated in numerous studies. Concerning blinks, one main function is to protect the eyes by maintaining the corneal tear film (Craig, 2002). However, longstanding work shows that most spontaneous blinks are influenced by cognitive tasks (Ponder & Kennedy, 1927). Studies reported that the probability of blinks is increased at the end of a stimulus evaluation processes in the visual (Fukuda, 2001; Wascher, Heppner, Möckel, Kobald, & Getzmann, 2015) as well as the auditory domain (Kobald, Wascher, Heppner, & Getzmann, 2019; Oh, Jeong, & Jeong, 2012) independent of ongoing sensory stimulation (Murali & Händel, 2021). Previous work further demonstrated that blink probability is modulated by the probability of stimulus occurrence in the auditory and the visual domain (Brych & Händel, 2020). Additionally, we could show in various studies using bistable visual and auditory stimuli, that the timing of blinks is linked to the perceptual interpretation of a stimulus. We find that a period without blinking reflects a phase in which a perceptual reinterpretation, in vision and audition, is facilitated while blink probability increases around the perceptual report (Brych et al., 2021; Murali & Händel, 2024). We interpreted the absence of blinks as marking a time in which attention is directed towards the sensory input whereas increased blinking marks an attentional disengagement from the sensory input because a viable interpretation already has been reached. It has been suggested that blinking might be associated with an attentional disengagement from visual stimulus (Ang & Maus, 2020; Nakano, Kato, Morito, Itoi, & Kitazawa, 2013; Salvi & Bowden, 2016). Our previous work extends this to sensory attention in general and distinguishes between sensory external attention and internally directed attention (Murali, 2023). Interestingly, rate and timing of blinks have been shown to be influenced by memory content (Karson et al., 1981; Ortega, Plaska, Gomes, & Ellmore, 2022) which shows influence on blinks that are independent of the presence of sensory input. Based on the above reviewed evidence one can assume that blinks are part of a domain-general mechanism. Accordingly, blinking can be relevant also during situations in which no visual processing is observed, such as in blind or during eyes closed.

Concerning saccades there is a similar influence of non-visual factors. Studies on memory have shown that rate and direction of saccades are influenced by memory content (Aagten-Murphy & Bays, 2019). Additionally, sensory influence does not seem to affect all saccades equally. Using an onset delay paradigm, it was shown that only a subpopulation of saccades was affected by a visual delay (Henderson & Pierce, 2008; Henderson & Smith, 2009). Concerning auditory influences, (Batten, 2018) reported a bi-modal fixation duration distribution, where one subpopulation was modulated by the sensory changes, whereas the other showed a rather invariable duration (245–250 ms). This suggests that not all saccades are influenced in its timing by sensory input (Hooge & Erkelens, 1998; Kolars, 1976). A possible explanation could be a sensory control mechanism of saccades which only takes over if necessary. If no control is expressed, the eyes might fall into a high frequency, small amplitude mode in which saccades are executed but without the goal to foveate. If a sensory driven gaze behavior is called for, the base rhythm is overruled by an active direction of saccades at variable amplitudes. Downbeat nystagmus is a condition where typically small amplitude saccades are involuntarily executed (following downward-directed slow eye movements) once the finely tuned motor control over the eyes is altered, such as in cerebellar patients (Hufner et al., 2007; Wagner et al., 2008). Such saccades, involuntarily executed at a relatively stable frequency, might further speak for the existence of a basic, sensory input-independent rhythm. The rhythmic muscle activity that is prevalent in blind and during eyes closed might be a representation of such a basic saccade rhythm. The role of saccadic rhythms outside vision is unclear.

### 13. Conclusion

This comparative study shows in a first step that blink rate, saccade rate, saccade amplitude and main sequences are comparable between EOG and video-based eye tracking in sighted during conditions of free viewing and without head restraint. This is true in a light and dark environment. After securing that EOG is a reliable tool to detect eye movements such as blinks and saccades in an unrestrained condition, we analyzed EOG data from sighted with eyes closed and congenitally blind participants. We find that blinks and saccades can still be detected during eyes closed and in blind. The detected eye movement show the expected features such as the main sequence and velocity profile, but are clearly distinguishable from events detected by the algorithm in surrogate data. When comparing to sighted participants with their eyes open, we find that the preferred saccade direction, rate and amplitude are different between conditions. The blink rate is not significantly different.

Overall, our study shows that EOG can be used to detect eye related movements in free moving setups, when the eyes are closed, and in sighted as well as congenitally blind participants. This suggests that eye movement are not exclusively introduced by visual input but should be viewed more holistically as part of the human body-cognition system.

### CRedit authorship contribution statement

**Barbara F. Händel:** Writing – original draft, Visualization, Supervision, Software, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Xinyu Chen:** Writing – review & editing, Project administration, Methodology, Investigation. **Maya Inbar:** Writing – review & editing, Project administration, Investigation, Data curation. **Flor Kusnir:** Writing – review & editing, Project administration, Investigation, Data curation. **Ayelet N. Landau:** Writing – review & editing, Supervision, Resources, Funding acquisition.

### Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Significance statement

Electrooculography (EOG) is a cost-effective and reliable tool for monitoring eye movements. Our findings demonstrate that EOG maintains its robustness even in minimally controlled environments, underscoring its applicability to novel research contexts and ecologically valid setups. This expands its utility for identifying biologically relevant markers. Furthermore, our study highlights the presence of blinks and saccadic muscle activity in sighted individuals with eyes closed and congenitally blind participants. These findings suggest that certain features of eye movements are not solely learned through visual feedback but are intrinsic to the human body-cognition system. This insight is particularly intriguing within the framework of active sensing, where sensory and motor processes are interlinked to facilitate perception and interaction with the environment.

## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.brainres.2025.149794>.

## Data availability

Data will be made available on request.

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